

PROBABILITY & STATISTICS

Previous Year Questions with Answers & Sample Problems

SECTION A: Previous Year Type Questions

Q1. Define expectation of a discrete random variable.

Answer:

$E[X] = \sum x p(x)$. It represents the average value of the random variable.

Q2. State and prove the linearity of expectation.

Answer:

$$E[X + Y] = E[X] + E[Y].$$

Proof follows from definition of expectation using joint distribution.

No independence assumption is required.

Q3. Define variance and derive $\text{Var}(X) = E[X^2] - (E[X])^2$.

Answer:

Variance is defined as $\text{Var}(X) = E[(X - \mu)^2]$.

Expanding the square gives $\text{Var}(X) = E[X^2] - (E[X])^2$.

Q4. State the condition for independence of two random variables.

Answer:

Two random variables X and Y are independent if:

$$p(X, Y) = p(X)p(Y).$$

Q5. Define covariance and show that $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.

Answer:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

Expanding the product gives $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.

Q6. Define Karl Pearson's coefficient of correlation.

Answer:

$$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y).$$

It measures the strength of linear relationship between two variables.

SECTION B: Sample Problems with Solutions

Problem 1:

A discrete random variable X has distribution:

X: 1, 2, 3

P(X): 0.2, 0.5, 0.3

Find $E[X]$ and $\text{Var}(X)$.

Solution:

$$\begin{aligned} E[X] &= 1(0.2) + 2(0.5) + 3(0.3) \\ &= 0.2 + 1 + 0.9 = 2.1 \end{aligned}$$

$$\begin{aligned} E[X^2] &= 1^2(0.2) + 2^2(0.5) + 3^2(0.3) \\ &= 0.2 + 2 + 2.7 = 4.9 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= 4.9 - (2.1)^2 \\ &= 4.9 - 4.41 \\ &= 0.49 \end{aligned}$$

Problem 2:

If $\text{Var}(X) = 4$, find $\text{Var}(3X + 2)$.

Solution:

$$\begin{aligned} \text{Var}(aX + b) &= a^2 \text{Var}(X) \\ &= 3^2 \times 4 \\ &= 9 \times 4 \\ &= 36 \end{aligned}$$

Problem 3:

Given $E[X] = 5$, $E[Y] = 3$, $E[XY] = 20$.

Find $\text{Cov}(X, Y)$.

Solution:

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= 20 - (5 \times 3) \\ &= 20 - 15 \\ &= 5 \end{aligned}$$

Problem 4:

If $\text{Cov}(X, Y) = 6$, $\sigma_X = 3$, $\sigma_Y = 4$,
find correlation coefficient.

Solution:

$$\begin{aligned} \rho &= \text{Cov}(X, Y) / (\sigma_X \sigma_Y) \\ &= 6 / (3 \times 4) \end{aligned}$$

$$= 6 / 12$$
$$= 0.5$$

Problem 5:

If X and Y are independent and $\text{Var}(X) = 2$, $\text{Var}(Y) = 5$,
find $\text{Var}(X + Y)$.

Solution:

For independent variables:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

$$= 2 + 5$$

$$= 7$$